Fake it till you make it: an examination of the US and English approaches to persona protection as applied to deepfakes on social media

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Introduction

The rise of social media has provided a platform for the use of celebrities' persona: name, voice, likeness and other persona features, by third parties. Brands often want to form an association with famous individuals due to the immense publicity value or advertising force embodied in the likeness or image of a well-known personality such as a movie actor or sports hero.[1] The commercial use of personas are mostly authorized in order to obtain the benefits of collaboration. There are, however, multiple instances of the less scrupulous use of an individual's persona without authorization.[2] Most notably, the emergence of 'deepfakes', a recent Artificial Intelligence (AI) development presents new challenges to regulations that question the applicability of existing law in the USA and the UK. Deepfakes involve a photograph or image manipulated to create a virtual representation of a person, which can be animated to portray the individual speaking or acting in a certain way.[3] For instance, a 'deepfake' video of Mark Zuckerberg promoting an art exhibition was created by two artists as part of a conceptual art project on AI.[4] As technology becomes more sophisticated, the range of uses of persona will potentially increase.

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This article

- The emergence of 'deepfakes', a recent Artificial Intelligence development, presents new challenges to regulations that question the applicability of existing law in the USA and the UK.
- As technology becomes more sophisticated, the range of uses of persona will potentially increase.
- While the use of deepfakes in advertising can be harmful to the consuming public who may be swayed under the guise of false celebrity endorsements, the ability to manipulate videos of public figures has further potentially grievous effects, particularly with regard to political influence.

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1 A Keel and R Nataraajan, 'Celebrity Endorsements and Beyond: New Avenues for Celebrity Branding' (2012) 29(9) Psychol Market 690, 690.

2 Some brands have shared false testimonials accompanied by the name and photograph of the individual. See C Milward, 'BBC News: 'I was horrified' Carol Kirkwood Furious over Unsettling Fake Advert Discovery' *Express* (London, 24 May 2019) (BBC TV presenter Carole Kirkwood's persona was used to endorse diet pills falsely. Her photograph was used as part of a fake news article which indicated that Kirkwood was leaving the BBC to establish a business which sells diet pills, much to Kirkwood's horror.). See also 'Teigen Warns Fans about Fake Endorsement Scam' (*Mama Mia*, 14 January 2019) <https://www.mamamia.com.au/chrissy-teigen-twitter-scam/> (Chrissy Tiegen's photograph was used by Keto Fit Premium, a diet pill company, in an advertisement which indicated Tiegen used their products to lose weight quickly after pregnancy) and 'Sandra Bullock and Ellen DeGeneres Team Up in Lawsuit to Stop Fake Ads' *Guardian* (London, 6 November 2019) (Sandra Bullock and Ellen DeGeneres' images are used without permission so frequently that they have launched a joint lawsuit against 100 entities for using their personas without permission to sell health products.)

3 A Wheeler, 'Ethics in Engineering and The Rise of Deepfakes' (*Engineering.com*, 20 August 2019) < https://www.engineering.com/DesignerEdge/DesignerEdgeArticles/ArticleID/19458/Ethics-in-Engineering-and-The-Rise-of-Deepfakes.aspx>.

4 Q Wong, 'Deepfake Video of Facebook CEO Mark Zuckerberg Posted on Instagram' (*CNet*, 11 June 2019) < https://www.cnet.com/news/deepfake-video-of-facebook-ceo-mark-zuckerberg-posted-on-instagram/>.

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under the guise of false celebrity endorsements, the ability to manipulate videos of public figures has further potentially grievous effects, particularly with regards to political influence.[5] To illustrate how deepfakes could contribute to misinformation and undermine the integrity of elections, a research institute manufactured a video which showed UK Prime Minister Boris Johnson falsely endorsing opposition leader Jeremy Corbyn in the run up to the 2019 General Election.[6] Similarly, the potential harm of synthetic media, even if it uses more traditional techniques, such as in a video of the Speaker of the US House of Representatives, Nancy Pelosi, which was digitally altered to make her appear inebriated, and published on Facebook.[7] Other examples include politicians such as Barack Obama and former Italian prime minister Matteo Renzi being made to appear to make crude comments or gestures towards fellow politicians.[8] At present, the majority of deepfakes have been used to impose the face of celebrities in pornography, thus raising the question of how this technology with potential for deceptive uses can be addressed under existing legislation.[9]

This article will explain that, despite jurisdictional differences, both US and UK claimants are likely to find that existing regulations will accommodate claims against deepfakes. However, the nature of harm caused by deepfakes necessitates responsibility on the part of social media platforms in detecting and removing these deceptive uses of persona.

What are deepfakes?

The term 'deepfake' was coined in 2017.[10] It is used to describe technology which manipulates photographic and audiovisual content to appear as though the individual featured is saying something or performing certain acts.[11] The term has assumed a negative connotation due to the dangers[12] it poses to those who may be the subject of a deepfake,[13] as well as the receiving public.[14] It has been used to portray celebrities in pornography,[15] and create advertisements which stir up controversy.[16] Deepfakes can be rapidly disseminated and cause mass misinformation, and have thus been likened to the fake news phenomenon.[17] The nature of deepfake technology necessitates an examination of the right of publicity and UK persona protection measures to determine what actions individuals can take to combat deepfakes. However, rectifying the social harms enumerated cannot be left to the aggrieved individuals as the current legal mechanisms provide inadequate tools.

United States right of publicity

The name, likeness and identity of a person enjoy broad protection under state law in the USA. The right of publicity was judicially established in the landmark decision of *Haelan Laboratories, Inc v Topps Chewing Gun Inc*[18] in 1953, after the 2nd Circuit recognized the unsuitability of applying the existing right of privacy to commercial portrayals of public figures. Subsequently, state-based rights of publicity emerged, some based on statute, others on common law or in some cases, both.[19] Generally, the right of publicity is strict liability and protects against commercial uses of persona. The following definition makes it clear that the right of publicity is primarily aimed at safeguarding the publicity value and commercial magnetism inherent in the name, likeness or identity of a person:

> The term 'right of publicity' has since come to signify the right of an individual, especially a public figure or a celebrity, to control the commercial value and exploitation of his name and picture or likeness and to prevent others

5 S Venkataramakrishnan, 'Can You Believe Your Eyes? How Deepfakes are Coming for Politics' *Financial Times* (24 October 2019) <https://www.ft.com/content/4bf4277c-f527-11e9-a79c-bc9acae3b654>.

6 BBC News, 'The Fake Video Where Johnson and Corbyn Endorse Each Other' (*BBC News*, 12 November 2019) <https://www.bbc.co.uk/news/av/technology-50381728/the-fake-video-where-johnson-and-corbyn-endorse-each-other>.

7 CBS News, 'Doctored Pelosi Video Highlights the Threat of Deepfake Tech' <https://www.youtube.com/watch?v=EfREntgxmDs>.

8 Venkataramakrishnan (n 5).

9 R Cellan-Jones, 'Deepfake Videos 'double in nine months'' *BBC* (London, 7 October 2019) <https://www.ft.com/content/4bf4277c-f527-11e9-a79c-bc9acae3b654>.

10 G Barber, 'Deepfakes are Getting Better, But They're Still Easy to Spot' (*Wired*, 26 May 2019) <https://www.wired.com/story/deepfakes-getting-better-theyre-easy-spot/?verso=true>.

11 R Chesney and D Citron, 'Deepfakes and the New Disinformation War' (2019) 98 Foreign Affairs 147.

12 R Chesney and D Citron, 'Deep Fakes: A Looing Challenge of for Privacy, Democracy, and National Security' (2019) 107 California Law Review (forthcoming).

13 B Marr, 'The Best (and scariest) Examples of AI-enabled Deepfakes' (*Forbes*, 22 July 2019) https://www.forbes.com/sites/bernardmarr/2019/07/22/the-best-and-scariest-examples-of-ai-enabled-deepfakes/.

14 Chesney and Citron (n 11) 147.

15 K Libby, 'This Bill Hader Deepfake Video is Amazing. It's Also Terrifying for Our Future' (*Popular Mechanics*, 13 August 2019) <https://www.popularmechanics.com/technology/security/a28691128/deepfake-technology/>.

16 O Schwart, 'You Thought Fake News Was Bad? Deep Fakes are Where Truth Goes to Die' *The Guardian* (London, 12 November 2018).

17 ibid.

18 (1953) 202 F. 2D 866 cert. denied 346 US 816, 98L. Ed. 343, 74 S. Ct. 26 (2nd Cir).

19 For example, California's right of publicity is both under the civil code at s 3344 and recognized at common law *White v Samsung Electronics America Inc* (1992) 971 F2d 1395.

from unfairly appropriating this value for their commercial benefit.[20]

In the USA, the breath of the right of publicity should present little difficulty in applying the law to the deepfakes in the commercial context of advertising. A series of cases in 1970s–1990s made it clear that persona extended to identity, which could include likeness, distinctive characteristics and voice. Cases such as *Ali v Playgirl Inc* (1978),[21] *Cher v Forum International Ltd* (1982)[22] and *Abdul-Jabbar v General Motors Corp* (1996)[23] demonstrated that nicknames and birth-names could be protected. The use of look-alikes was also redressed through the right of publicity, or false endorsement in *Onassis v Christian Dior New York Inc* (1984),[24] *Allen v National Video Inc* (1985)[25] and *Newcombe v Adolf Coors Co* (1998).[26] The use of voice and soundalikes were considered in *Shaw v Time-Life Records* (1981),[27] *Midler v Ford Motor Co* (1988)[28] and *Waits v Frito Lay Inc* (1992).[29] Identity was examined in *Motschenbacher v RJ Reynolds Tobacco Co* (1974),[30] *Carson v Here's Johnny Portable Toilets Inc* (1983),[31] *Vanna White v Samsung Electronics America Inc* (1992)[32] and *Wendt & Ratzenberger v Host International Inc & Paramount Pictures Corp* (1997),[33] and found to include personal belongings, catchphrases and robotic likenesses, respectively. As deepfakes are life-like portrayals of individuals, they are likely to be within the scope of the right of publicity (where recognized).

While commercial uses of persona are clearly within the remit of the right of publicity, there are specific exceptions for news reporting, public interest, sports broadcast and political campaigns.[34] Occupying a middle ground, non-commercial works such as movies and video games may avail of defences.[35] Three tests have evolved to determine the boundaries of First Amendment protection that may be granted to non-commercial works: The Missouri Predominant Use test,[36] mainly confined to its state of creation,[37] the *Rogers* Test used by the sixth circuit[38] and the most popular Transformative Use test developed by the California Supreme Court in the 2001 case of *Comedy III Productions Inc v Gary Saderup*.[39]

According to the 3rd Circuit in *Comedy III,* when an artist is faced with a right of publicity challenge to his or her work, they may raise as affirmative defence that the work is protected by the First Amendment inasmuch as it contains significant transformative elements, and the value of the work does not derive primarily from the celebrity's fame.[40] The court in *Comedy III* considered 'whether the new work merely supersedes the objects of the original creation, or instead adds something new, with a further purpose or different character, altering the first with new expression, meaning or message'.[41] In that case, a lithograph on T-shirts was found to infringe the plaintiffs' right of publicity because they did not contain creative elements that would sufficiently transform the work into protectable expressive free speech. More recently, individuals have successfully claimed that their right of publicity was infringed by life-like portrayals in video games. The US Supreme Court confirmed that video games are expressive works in 2011, subject to First Amendment protection.[42]

It is worth examining the treatment of video games by courts as the technology deployed seeks to create life-like portrayals of individuals. While the portrayals are not as realistic as the deepfake technology allows, the cases can be indicative as to how deepfakes which are used in non-commercial settings may be treated. *Re NCAA Student-Athlete Name & Licensing*

20 *Estate of Presley v Russen* (D.N.J. 1981) 513 F Supp. 1339, 1353.
21 447 FSupp 723.
22 213 USPQ 96 (CD Cal).
23 85 F 3d 407.
24 122 Misc 2d 603, 427 NYS2d 254.
25 610 F Supp 612 (SDNY).
26 157 F 3d 686 (9th Cir).
27 38 NY 2d 201 (1975).
28 849 F 2d 460 (9th Cir).
29 978 F 2d 1093 (9th Cir).
30 498 F2d 821 (9th Cir).
31 98 F2d 831.
32 971 F2d 1395.
33 125 F 3d 806, 44 USPQ 2d 1189 (9th Cir).
34 M Redish and K Shust, 'The Right of Publicity and the First Amendment in the Modern Age of Commercial Speech' (2014) 56 Wm & Mary L Rev 1443.
35 Czarnota, 'The Right of Publicity in New York and California: A Critical Analysis' (2012) 19 Jeffrey S. Moorad Sports LJ 481.
36 *Doe v TCI Cablevision* (Mo 2003) 110 SW 3d 363, 370, 31 Media L Rep (BNA) 2025, 67 USPQ 2d 1604.
37 T McCarthy, *Rights of Publicity and Privacy* (Thomson Reuters 2018) s 8.23.
38 *Rogers v Grimaldi* (2d Cir 1989) 875 F 2d 994, 999, 16 Media L Rep (BNA) 1648, 10 USPQ 2d 1825.
39 (Cal) 21 P 3d 797. It should be noted that this case concerned statutory post-mortem publicity rights held by registered owner of all rights of deceased celebrities known as The Three Stooges. Not all jurisdictions provide post-mortem protection. Notably, there is no post-mortem publicity protection in New York.
40 *Comedy III*, ibid [50].
41 ibid [42].
42 Like the protected books, plays and movies that preceded them, video games communicate ideas—and even social messages—through many familiar literary devices (such as characters, dialogue, plot and music) and through features distinctive to the medium (such as the player's interaction with the virtual world). That suffices to confer First Amendment protection. *Brown v Entertainment Merchants Association* (2011) 564 US 786 [790].

Litigation[43]and *Davis v Electronic Arts*,[44] the likenesses and names of college athletes were used in reality sports video games. There was no question that the characters in the video games were based on the athletes as the purpose of the games was to allow players to use avatars of the athletes. The video game failed the 'transformative use' test because it recreated the plaintiffs insofar as possible. A similar outcome was achieved in *No Doubt v Activision Publishing, Inc*, where the court opined that the depictions of the members of the band were 'pure mimicry' (the case was settled after a second appeal).[45] This litigation has clarified that life-like portrayals of individuals in expressive works will be subject to application of the right of publicity, indicating that individuals depicted in deepfakes should not face any legal barriers to potential claims.[46]

UK piecemeal approach to persona protection

A more conservative approach to the protection of publicity value and commercial magnetism embodied in the name, likeness or photograph of an individual is found in UK law, which takes a piecemeal approach to protection. One relevant cause of action is defamation, governed by the common law and the Defamation Act 2013.

Defamation was used to control an unauthorized advertisement in *Tolley v IS Fry and Sons, Limited*.[47] In this case, a chocolate manufacturer used, without consent, a caricature of the plaintiff, a prominent amateur golfer, depicting him as playing golf with a packet of their chocolate protruding from his pocket. The court opined that the unauthorized use of plaintiff's likeness carried the innuendo that the plaintiff compromised his status as an amateur golfer. More recently, in 2016, Martin Lewis,[48] journalist, TV presenter and financial adviser, initiated High Court proceedings against Facebook for hosting unauthorized advertisements using his persona. He dropped his suit after Facebook agreed to donate £3 million to Citizens Advice for a 'Scams action' project, and a UK scam advertisement reporting tool on the platform.[49]

Defamation does not protect the individual against unauthorized use of persona per se. Rather, it is conditional on the portrayal causing serious harm to the reputation of the claimant.[50] The position of defamation is best explained by Bains. He points out that:

> Defamation law is however not the best means by which to safeguard a celebrity's personality since it does not protect against appropriation and exploitation of one's personality, but only against the criticism and ridicule of their personality.[51]

While defamation is more limited than the right of publicity, it can serve as a useful tool against non-commercial uses of deepfakes. This is particularly so in a context where there is a lowering of someone's image in the eyes of the digital public.

Another avenue of protection for a celebrity's publicity value is the tort of passing off. A passing off claim requires three elements: goodwill, misrepresentation and damage.[52] While goodwill will usually be easy for celebrities and sportspeople to establish, the real hurdle is posed by misrepresentation. Unlike the strict liability US right of publicity, the misrepresentation element requires that consumers be deceived into thinking that the individual endorsement or approved of the use of her persona in connection with the product.

Misrepresentation used to be an insurmountable hurdle in passing off cases due to the 'common field of activity' requirement, which meant that the claimant had to be in the same industry as the defendant. The first case which attempted to rely on passing off to protect personal image was *McCulloch v Lewis A May* (*Uncle Mac* case).[53] In this case, a children's radio host contested the use of his moniker, 'Uncle Mac', on cereal. Passing off was not established, as the court held that there was no misrepresentation due to a lack of common field of activity between the claimant and defendant. That is, Uncle Mac was not in the business of making cereals so that the relevant consumer would not think that the cereal was produced by him. Subsequently, the common field of activity proved fatal in *Wombles Ltd v Wombles Skips Ltd*.[54] The names of fictional characters, the Wombles, were used by a

43 (9th Cir 2013) 724 F 3d 1268 ("Keller").

44 (9th Cir 2015) 775 F 3d 1172.

45 (2011) 192 Cal.App 4th 1018, 122 Cal Rptr 3d 397, 400 [410].

46 T Kadri, 'The Right of Publicity Goes 2 – 0 Against Freedom of Expression' (2014) 112(8) Mich L Rev 1519, 1524–25, *Keller v Elec Arts Inc (In re NCAA Student-Athlete Name & Likeness Licensing Litig)* (9th Cir 2013) 724 F 3d 1268; *Hart v Elec Arts, Inc* (3d Cir 2013) 717 F 3d 141. (3d Cir 2013).

47 [1931] AC 333.

48 T Bradshaw, 'Facebook Settles Martin Lewis Lawsuit over "Scam Ads"' *Financial Times* (London, 23 January 2019).

49 Citizens Advice, 'Report a Scam'<https://www.citizensadvice.org.uk/consumer/scams/reporting-a-scam/>

50 s 1 Defamation Act 2013.

51 S Bains, 'Personality Rights: Should the UK Grant Celebrities a Proprietary Right in Their Personality? Part 1 (2007) 18 Ent L Rev 164, 167.

52 *Reckitt & Colman Ltd v Borden Inc* [1990] 1 All ER 873.

53 [1947] 2 All ER 845.

54 (1975) FSR 488 (1977) RPC 99.

company selling skips. No passing off was found to be present because the relevant consumer would not think the skips originated from the producers of the TV show. Soon after, ABBA unsuccessfully claimed against use of photographs on badges, pillow slips and T-shirts as the court was not convinced that the relevant consumer would think there was a link.[55] Similarly, in *Harrison and Starkey v Polydor*, the Beatles claim for use of pictures on an album cover failed as the court said no one would infer that they approved or sponsored the album.[56]

The common field of activity requirement was finally abandoned by the court *in LegoSystems A/S v Lego M Lemelstrich*[57] in 1983. In *Mirage Studios v Counter-Feat Clothing Company Limited*, the court found that passing off was successful as the relevant consumer would think that T-shirts depicting the ninja turtles were 'official' in the sense that the manufacturer had secured a licence.[58] In this case the court noted that the common field of activity requirement had been formerly discredited. However, it can be suggested that, even if the common field of activity requirement was imposed, it is likely that a finding of passing off would have subsisted as the court recognized that merchandising was a recognized industry.[59]

The merchandising of fictional characters has also been addressed more recently in the case of *Hearst Holdings v AVELA*.[60] The holders of the rights to 'Betty Boop' successfully claimed passing off against unauthorized merchandise.[61] It is worth noting that Birss J observed potential differences between cases concerning fictional characters and real people. He noted that:

> For example real people can be photographed by anyone and so the public are used to seeing images of famous people in contexts far removed from endorsement or merchandising. The same may not be the case for invented characters.[62]

Birrs J did not go as far as to say that fictional characters are more likely to receive protection than real individuals, but the contextualization of how the public is exposed to these different types of claimants could be indicative of such. This said, the creation of fake merchandise could be considered a precursor to deepfake technology. It was easier to reproduce an image of a fictional character than it was to create a video falsely depicting an individual until the dawn of this new technology which removes previous technological barriers.

While no claims of passing off have been brought against deepfakes, celebrities have found success by using the tort of passing off to protect against unauthorized advertising and merchandising uses. In the case of *Irvine v Talksport*, Eddie Irvine successfully claimed passing off when TalkSport radio used a photograph of the famous race car driver without his permission. The photograph was edited to make it appear that Irvine was listening to a radiolabelled 'TalkSport'. The court acknowledged that Irvine had licensed his persona to many brands, and that consumers were likely to think he had similarly endorsed this product. Similarly, Rihanna successfully sued Topshop for using a photograph of her on a T-shirt in *Fenty and others v Arcadia Group Brands Ltd (trading as Topshop) and another*.[63] While Topshop was in possession of the necessary copyright licence for the photograph, Rihanna argued that consumers would be misled into thinking she approved the product. A wealth of evidence including a past collaboration, the nature of the photograph as part of her 'Talk that Talk' album photo shoot, Topshop's tweet about her in their Oxford Street flagship store, as well as Rihanna's position as a fashion industry leader, contributed to a finding of misrepresentation.

While *Irvine* and *Fenty* are the only cases where celebrities have attempted to use passing off to protect their persona in the last 20 years, it is suggested that misrepresentation should be easier than ever to prove since the use of persona in a deepfake is so life-like. Advertisers are likely to use deepfakes to send a positive message about their brand, thus conveying endorsement and satisfying the misrepresentation element. Arguably, the ability to manipulate 'deepfakes' make them even more amendable to passing off claims than the uses of mere photographs in the *Irvine* and *Fenty*. While the approach to persona protection in the UK is not as comprehensive as the USA, the nature of 'deepfake' technology fits more comfortably into available causes of action, namely defamation and passing off, than any previously conceivable use of persona.

55 *Lyngstad v Anabas* [1977] FSR 62.
56 [1977] FSR 1.
57 [1983] FSR 155.
58 [1991] FSR 145.
59 ibid [148].
60 [2014] EWHC 439 (Ch).
61 The claimants also succeed under their claim for trade mark infringement.
62 ibid [71].
63 [2013] EWHC 1945 (Ch), [2015] EWCA Civ 3.

Personal causes of action as an inadequate tool of redress against the harm posed by deepfakes

It has been argued that both the USA and UK have causes of action which may be used to address deepfakes. However, the rights to take action against unauthorized deepfakes vest in the individual portrayed. The usefulness of these causes of actions is thus limited as they do not adequately address the potential harms posed by deepfakes, namely personal and public harm. Public harm can result from deepfakes due to the potential to sway public opinion. This potential is even greater than that of fake news, a concept which was popularized during the 2016 US election.[64] The revelation of the Cambridge Analytica scandal demonstrated how fake news can be used for political influence.[65] Fake news stories were deliberately disseminated in order to sway social media users to vote a certain way for the election.[66] Deepfakes are potentially more damaging because they appear to be legitimate. They can be used to create a fake video of a political figure which could then be quickly disseminated through social media as well as addressed by traditional media. It is unlikely that web users will be able to discern when a deepfake is being used, and traditional media would also face difficulties unless they are able to avail of relevant technology. Even with the availability of relevant technology, it is questionable whether traditional media would undertake the task of evaluating the legitimacy of a video. The risks of deepfakes were highlighted by Jordan Peele and BuzzFeed when they collaborated to create a deepfake of Barak Obama warning about the dangers of deepfake technology.[67]

The risks associated with deep fakes and the potential for rapid dissemination can be extrapolated from data collected on fake news. A research project by MIT recognized that 'new social technologies, which facilitate rapid information sharing and large-scale information cascades, can enable the spread of misinformation (ie information that is inaccurate or misleading)'.[68] The study found that it took six times as long for accurate information to reach 1500 people as compared to fake news.[69] One of the reasons for the disparity in disseminations is that novel information is more likely to be shared.[70] The entrenchment of fake news can be likened to the potential for deepfakes to spread false information. The public nature of this harm necessitates that steps are taken to detect and remove deepfakes. It should not be left to the responsibility of the individual portrayed to be the gatekeeper of information, particularly as the actions discussed above are designed to provide personal redress.

Even the availability of personal redress is insufficient considering the nature of deepfakes which can cause serious reputational damage. While not a deepfake, a video of the Speaker of the US House of Representatives, Nancy Pelosi, was edited to make her appear inebriated, with slurred speech.[71] Deepfakes can similarly be used as they are not simply a matter of making false statements about an individual or associating an individual with an unwanted product. Rather, deepfakes make it appear that the individual has spoken statements or acted in a certain manner. This technology can thus be used to make it appear that an individual has engaged in hate speech, violence and in pornographic acts.[72] The potential manipulation of persona exceeds anything that was previously fathomable, and the harm to the reputation, dignity and integrity to the individual would be unprecedented. While defamation in the UK and the right of publicity in the USA could be deployed against deepfakes, it might be difficult to track down the uploader of the work, especially as the work may go viral and be shared across platforms. Even if a potential defendant could be identified, the time between identification and commencing legal action creates irreparable damage.

Social media platforms are in a better position than the individual to monitor unauthorized uses of persona. As mentioned above, Facebook has launched an anti-scam tool to tackle false advertisements. Pornhub, Twitter and Gfycat have banned pornography using deepfakes.[73] Similar measures should be implemented by Instagram, YouTube and other social media

64 K Hall, 'Deepfake Videos: When Seeing Isn't Believing' (2018) 27 Cath U J L & Tech 51, 55.

65 'Cambridge Analytica a Year On: "A Lesson in Institutional Failure"' *The Guardian* (London, 17 March 2019).

66 'Cambridge Analytica Planted Fake News' *BBC* (London, 20 March 2018); M Blitz, 'Lies, Line Drawing and Deep Fake News' (2018) 71 Okla L Rev 59, 64–65.

67 R Whitman, 'Buzzfeed Created a 'Deepfake' Obama PSA Video' *Extreme Tech* (18 April 2018) <https://www.extremetech.com/extreme/267771-buzzfeed-created-a-deepfake-obama-psa-video>.

68 S Vosoughi, D Roy and S Aral, 'The Spread of True and False News Online' (2018) 359(6380) Science 1146, 1146.

69 ibid.

70 'Although we cannot claim that novelty causes retweets or that novelty is the only reason why false news is retweeted more often, we do find that false news is more novel and that novel information is more likely to be retweeted.' Vosoughi, Roy and Aral (n 68) 1150.

71 C Warzel, 'The Fake Nany Pelosi Video Hijacked Our Attention. Just as Intended' *The New York Times* (New York, 26 May 2019).

72 D Harris, 'Deepfakes: False Pornography is here and the Law Cannot Protect You' (2018-2019) 17 Duke L & Tec Rev 99.

73 A Hern, '"Deepfake" Face-swap Porn Videos Banned by Pornhub and Twitter' *The Guardian* (London, 7 Feburary 2018).

platforms to monitor the use of deepfake technology. Early identification and removal of material are essential to prevent the harms enumerated above from occurring. While the harm posed by advertising can be addressed in this way, it is arguably more pressing to ensure that unauthorized non-commercial uses are regulated as these have the potential to sway public opinion and cause grievous harm to the individual portrayed. It is proposed that content moderation rules of social media platforms should specifically provide for a category on the abuse of a person's identity via deepfakes.

Content moderation rules would be of little value without enforcement. This means that platforms need access to technology which can detect deepfakes. Facebook, Microsoft and Partnership on AI have launched an initiative to develop such technology.[74] The 'Deepfake Detection Challenge' is a partnership among the aforementioned organizations and leading universities include Cornell, MIT and Oxford.[75] While this effort is to be welcomed, there are inevitable limits on the ability of technology to quickly detect, assess and remove large volumes of content. Klonick details the process of content moderation,[76] which entails reactive moderation when material is brought to the attention of the moderator, and proactive moderation which occurs when the moderation actively seeks out content.[77] Moderation may be automated via software designed for this purpose or manually done by humans trained in content moderation.[78] Moderation is an imperfect process, and the ability of content to go viral within a matter of moments means that there will inevitably be content which is viewed before it can be removed. A harrowing example is that of the live footage broadcasted by the New Zealand Christchurch massacre in 2019.[79] While Facebook removed the video upon being notified, the video had already been viewed numerous times and even supported by individuals sympathetic to the shooter.

Even with the relevant technology, content moderation is challenging because it has the potential to impinge on free speech. As noted by Langvardt: 'Such a system [removal of content' whether is it condemned as "censorship" or accepted as "content moderation," sits in tension with an American free speech tradition'.[80] This is not just a US issue, as freedom of expression is also protected under the European Union Charter of Human Rights and the European Convention on Human Rights. While this article does not attempt to resolve the conflict between ontent moderation and the balance to be struck with freedom of speech, it is worth noting that any new technology is likely to face challenges in effectiveness and deployment.

Conclusion

In both the UK and USA, deepfakes are likely to be addressed by current law. In the UK, passing off, while more complex than the right of publicity, could be helpful to address deepfakes where they are used in a way that conveys endorsement. For non-endorsement uses, defamation might be grounds for legal action. In the USA, the right of publicity can be deployed. However, these causes of action inadequately address the potential of deepfakes to sway public opinion, damage the reputation of the individual depicted, and impinge on privacy and dignity of the individual if used in pornography, for example. It may be challenging to track down the uploader, and the delay between commencing legal action and receiving an injunction can be too late if the deepfake goes viral. The potential velocity and amplification of deepfakes on the web allows for nefarious uses of this relatively new technology.

In this context of the digital era, social media platforms have a social responsibility to monitor and take down deepfakes. Developing specific provisions targeted at deepfakes could be a useful first step. For instance, Twitter has devised specific provisions on 'synthetic and manipulated media' to combat misinformation and protect the integrity of discourse on the platform.[81]

While the creation of rules in this regard is useful, there is also a need for technology to facilitate content moderation. As deepfake videos become increasingly difficult to distinguish from real footage, news organizations such as Reuters are conducting their own

74 D Fuscaldo, 'Facebook, Microsfot and Academia Launch New Challenge to Develop Technology to Spot Deepfakes' (*Interesting Engineering*, 5 September 2019).

75 ibid.

76 K Klonick, 'The New Governors: The People, Rules and Processes Governing Online Speech' (2018) 131 Harv L Rev 1599, 1630–62.

77 ibid 1635.

78 ibid 1635.

79 J Abbruzzese and B Zadronzy, 'Streamed to Facebook, Spread on YouTube: New Zealand Shooting Video Circulates Online Despite Takedowns' *NBC News* (15 March 2019) <https://www.nbcnews.com/tech/tech-news/streamed-facebook-spread-youtube-new-zealand-shooting-video-circulates-online-n983726>.

80 K Langvardt, 'Regulating Online Content Moderation' (2018) 106 Geo LJ 1353, 1359.

81 See <https://blog.twitter.com/en_us/topics/company/2019/synthetic_manipulated_media_policy_feedback.html>

deepfakes experiments to stay one step ahead of the fake news disseminators.[82] Detection of deep fakes is still largely an open research problem.[83] Deepfakes are already approaching the limits of human perception to discriminate. Automated techniques have focused on heuristics such as blinking patterns or incongruities in perspective and shading. It is unlikely that these will remain effective for very long. The arms race will continue for the foreseeable future, and as such detecting deepfakes may well remain an aspirational goal rather than something that can be relied upon as a solution.

The 'Deepfake Detection Challenge' initiative is to be welcomed, but the challenges posed by content moderation mean that some nefarious uses of deepfake technology will inevitably slip through the cracks. It remains to be seen how responsive social media platforms will be in regulating the use of deepfakes.

82 H Baker, 'Making a 'deepfake': How Creating Our Own Synthetic Video Helped Us Learn to Spot One' (*Reuters*, 11 March 2019) <https://www.reuters.com/article/rpb-deepfake/making-a-deepfake-how-creating-our-own-synthetic-video-helped-us-learn-to-spot-one-idUSKBN1QS2FO>.

83 Yuezun Li and Siwei Lyu, 'Exposing DeepFake Videos By Detecting Face Warping Artifacts' <http://openaccess.thecvf.com/content_CVPRW_2019/papers/Media%20Forensics/Li_Exposing_DeepFake_Videos_By_Detecting_Face_Warping_Artifacts_CVPRW_2019_paper.pdf>. Deepfakes typically use Generative Adversarial Networks, which consists of a generator and discriminator component in competition with each other. The discriminator learns, over time, to effectively distinguish between media created by the generator and media supplied from a real-world data set. The technique for creating deep-fakes, therefore, fundamentally narrows the scope of techniques for detecting them, because any vulnerability to detection can be added to the discriminator and trained out of the creation process.